

DSA Master Sheet											
Sr No.	Sub Sr No.	Topic	Subtopic	Problem Name	Link	Difficulty	Status	Notes	Revisit		
PHASE 1 : FUNDAMENTALS & LINEAR DATA STRUCTURES											
1	1	Arrays	Array Basics	Find Element at a Given Index	Link	☆☆☆☆	Not Attempted		Not Attempted		
2	2			Min and Max in Array	Link	☆☆☆☆	Tried (couldn't think of logic)		Revisit		
3	3			Sum of Array	Link	☆☆☆☆	Logic Done (couldn't code)		Can Revisit		
4	4			Sum of Digits	Link	☆☆☆☆	Code Done (but has error)		No Revisit Required		
5	5			Check If Array is Sorted	Link	☆☆☆☆	Done				
6	6			Alternates In Array	Link	☆☆☆☆	Done				
7	7		Core Manipulations	Remove Duplicates from Array	Link	☆☆☆☆					
8	8			Second Largest in Array	Link	☆☆☆☆					
9	9			Reverse an Array	Link	☆☆☆☆					
10	10			Missing Number	Link	☆☆☆☆					
11	11			Segregate 0s and 1s	Link	☆☆☆☆					
12	12			Maximum Consecutive Ones	Link	☆☆☆☆					
13	13			Palindromic Array	Link	☆☆☆☆					
14	14			Move Zeroes to End	Link	☆☆☆☆					
15	15		Intermediate Problems	Sort array with 0's 1's and 2's (Dutch Flag)	Link	☆☆☆☆					
16	16			Equilibrium Point	Link	☆☆☆☆					
17	17			Reverse Integer	Link	☆☆☆☆					
18	18			Leaders in Array	Link	☆☆☆☆					
19	19			Increasing Array	Link	☆☆☆☆					
20	20			Rearrange Array Elements by Sign	Link	☆☆☆☆					
21	21			Rotate Array by One	Link	☆☆☆☆					
22	22			Majority Element I (Boyer - Moore)	Link	☆☆☆☆					
23	23			Advanced Problems	Rotate Array by K steps	Link	☆☆☆☆				
24	24				Wiggle Sort II	Link	☆☆☆☆				
25	25		Majority Element II		Link	☆☆☆☆					
26	26		Best Time to Buy and Sell Stock		Link	☆☆☆☆					
27	27		Next Permutation		Link	☆☆☆☆					
28	28		Maximum Value Of Expression		Link	☆☆☆☆					
29	29		First Missing Positive		Link	☆☆☆☆					
30	1	2D Arrays	Matrix Basics	Sum of elements in a matrix	Link	☆☆☆☆					
31	2			Count Number Of Zeroes	Link	☆☆☆☆					
32	3			Count Negative Numbers in a matrix	Link	☆☆☆☆					
33	4			Matrix Diagonal Sum	Link	☆☆☆☆					
34	5		Matrix Operations	Addition of Two Square Matrix	Link	☆☆☆☆					
35	6			Multiply Matrices	Link	☆☆☆☆					
36	7			Transpose Matrix	Link	☆☆☆☆					
37	8		Advanced Matrix	Spiral Matrix	Link	☆☆☆☆					
38	9			ZigZag Matrix	Link	☆☆☆☆					
39	10			Rotate Matrix (90°)	Link	☆☆☆☆					
40	11			Determine Whether matrix can be obtained by rotation	Link	☆☆☆☆					
41	12			Set Matrix Zeroes	Link	☆☆☆☆					
42	1	Basic Maths	Number Theory Basics	Check if a number is Armstrong	Link	☆☆☆☆					
43	2			Print all Divisors of a Number	Link	☆☆☆☆					
44	3			Check if a number is Prime	Link	☆☆☆☆					
45	4			GCD / HCF of 2 numbers	Link	☆☆☆☆					
46	5			Prime Factorisation of a Number	Link	☆☆☆☆					
47	6			Count Primes in range L to R	Link	☆☆☆☆					
48	1	Strings	String Fundamentals	Maximum Occuring Character	Link	☆☆☆☆					
49	2			Remove Spaces	Link	☆☆☆☆					
50	3			Print first letter of every word in the string	Link	☆☆☆☆					
51	4			Remove Consecutive Characters	Link	☆☆☆☆					
52	5			Valid Palindrome	Link	☆☆☆☆					
53	6			Valid Anagram	Link	☆☆☆☆					
54	7			Isomorphic Strings	Link	☆☆☆☆					
55	8		String Manipulation	Delete Characters To Make Fancy String	Link	☆☆☆☆					
56	9			Reverse Words in a String	Link	☆☆☆☆					
57	10			String to integer (atoi)	Link	☆☆☆☆					
58	11			Roman to integer	Link	☆☆☆☆					
59	12			Rotate String	Link	☆☆☆☆					
60	13			Longest common prefix	Link	☆☆☆☆					
61	14		Advanced Problems	Longest palindromic substring	Link	☆☆☆☆					
62	15			Multiply Two Strings	Link	☆☆☆☆					
63	16			Maximum Nesting Depth of the Parentheses	Link	☆☆☆☆					
64	17			Beauty Of All substrings	Link	☆☆☆☆					
PHASE 2 : SEARCHING ALGORITHM											
65	1		BS on 1D Array Basics	Search X in sorted array	Link	☆☆☆☆					
66	2			Lower Bound	Link	☆☆☆☆					
67	3			Upper Bound	Link	☆☆☆☆					
68	4			Search insert position	Link	☆☆☆☆					
69	5			Floor In Sorted Array	Link	☆☆☆☆					
70	6			Ceil In Sorted Array	Link	☆☆☆☆					
71	7			Guess Number Higher or Lower API	Link	☆☆☆☆					
72	8			First 1 in a Sorted Binary Array	Link	☆☆☆☆					
73	9			Kth Missing Positive Number	Link	☆☆☆☆					
74	10			Find minimum in Rotated Sorted Array	Link	☆☆☆☆					

[illegible]

[illegible]

231	24		Graph-Like Traversals in Trees	Binary Tree Maximum Path Sum	Link	★★★★★				
232	25			Path Sum III	Link	★★★★★				
233	26			Minimum time taken to burn the BT	Link	★★★★☆				
234	27			All Nodes Distance K in Binary Tree	Link	★★★★★				
235	28			Step-By-Step Directions From a Binary Tree Node to Another	Link	★★★★★				
236	29			Merge Two Binary Trees	Link	★★★★☆				
237	30			Children Sum Property in a Binary Tree	Link	★★★★☆				
238	31			Construct Binary Tree from Preorder and Inorder Traversal	Link	★★★★★				
239	32			Construct Binary Tree from Inorder and Postorder Traversal	Link	★★★★☆				
240	33			Flatten Binary Tree to Linked List	Link	★★★★★				
241	34			Serialize and Deserialize Binary Tree	Link	★★★★★				
242	1	Binary Search Trees	Core Properties & Search	Search in a Binary Search Tree	Link	★★★★☆				
243	2			Find Minimum/Maximum in BST	Link	★★★★☆				
244	3			Insert into a Binary Search Tree	Link	★★★★☆				
245	4			Floor and Ceil in a BST	Link	★★★★☆				
246	5			Inorder Successor in BST	Link	★★★★★				
247	6			Lowest Common Ancestor of a Binary Search Tree	Link	★★★★☆				
248	7			Validate Binary Search Tree	Link	★★★★★				
249	8			Kth Smallest Element in a BST	Link	★★★★★				
250	9			Standard Modifications & Deletions	Delete Node in a BST	Link	★★★★★			
251	10			Two Sum IV - Input is a BST	Link	★★★★★				
252	11	Construction & Advanced Operations	Convert Sorted Array to Binary Search Tree	Link	★★★★☆					
253	12		Construct BST from Preorder Traversal	Link	★★★★★					
254	13		Balance a Binary Search Tree	Link	★★★★★					
255	14		Merge Two Balanced Binary Search Trees	Link	★★★★★					
256	15		Recover Binary Search Tree	Link	★★★★★					
257	16		Largest BST in a Binary Tree	Link	★★★★★					
258	1	Tries	Implementation & String Search	Implement Trie (Prefix Tree)	Link	★★★★☆				
259	2			Design Add and Search Words Data Structure	Link	★★★★★				
260	3			Longest Word with All Prefixes	Link	★★★★★				
261	4			Number of Distinct Substrings in a String	Link	★★★★☆				
262	5		Word Search II	Link	★★★★★					
263	6		Bitwise Tries (Advanced)	Maximum XOR of Two Numbers in an Array	Link	★★★★★				
264	7			Maximum XOR With an Element From Array	Link	★★★★★				
PHASE 7 : CORE DATA STRUCTURES AND TECHNIQUES										
265	1	Hashmaps	Hashmap Fundamentals	Contains Duplicate	Link	☆☆☆☆				
266	2			Valid Anagram (Hashmap Approach)	Link	☆☆☆☆				
267	3			Unique Number of Occurrences	Link	☆☆☆☆				
268	4			Find distinct elements / Find the Frequency	Link	☆☆☆☆				
269	5			Two Sum	Link	☆☆☆☆				
270	6		Intersection of Two Arrays	Link	☆☆☆☆					
271	7		Count Number of Pairs With Absolute Difference K	Link	☆☆☆☆					
272	8		Design HashMap	Link	☆☆☆☆					
273	9		Intermediate Hashmap Problems	Group Anagrams	Link	☆☆☆☆				
274	10			Longest Consecutive Sequence	Link	☆☆☆☆				
275	11	Subarray Sum Equals K		Link	☆☆☆☆					
276	12	Contiguous Array / Largest subarray with 0 sum		Link	☆☆☆☆					
277	13	Count subarrays with given XOR		Link	☆☆☆☆					
278	14	Advanced / Multi Concept	Subarray Sums Divisible by K	Link	☆☆☆☆					
279	15		Continuous Subarray Sum	Link	☆☆☆☆					
280	16		Maximum Size Subarray Sum Equals k	Link	☆☆☆☆					
281	17		Top K Frequent Elements	Link	☆☆☆☆					
282	18		LRU Cache (Least Recently Used)	Link	☆☆☆☆					
283	19		LFU Cache (Least Frequently Used)	Link	☆☆☆☆					
284	1	Heaps / Priority Queues	Heap Fundamentals	Check if an array represents a min heap	Link	☆☆☆☆				
285	2			Convert Min Heap to Max Heap	Link	☆☆☆☆				
286	3			Implement Min/Max Heap		☆☆☆☆				
287	4		The Top-K Pattern	Last Stone Weight	Link	☆☆☆☆				
288	5			Kth Largest Element in an Array	Link	★★★★★				
289	6			K Closest Points to Origin	Link	★★★★★				
290	7			Sort a K Sorted Array / Nearly Sorted Algorithm	Link	★★★★★				
291	8		Merging & Combinations	Top K Frequent Elements (Heap Approach)	Link	★★★★★				
292	9			Minimum Cost of Ropes	Link	★★★★★				
293	10			Merge K Sorted Lists	Link	★★★★★				
294	11	Advanced / Two-Heaps / Scheduling	Kth Smallest Element in a Sorted Matrix	Link	★★★★★					
295	12		Task Scheduler	Link	★★★★★					
296	13		Reorganize String	Link	★★★★★					
297	14		Minimum Number of Refueling Stops	Link	★★★★★					
298	15		Find Median from Data Stream	Link	★★★★★					
299	1	Prefix Sum	1d Prefix Sum	Running Sum of 1D Array	Link	☆☆☆☆				
300	2			Find the Highest Altitude	Link	☆☆☆☆				
301	3			Find Pivot Index	Link	☆☆☆☆				
302	4			Range Sum Query - Immutable	Link	☆☆☆☆				
303	5			Product of Array Except Self	Link	★★★★★				
304	6		2d Prefix Sum	Corporate Flight Bookings	Link	★★★★★				
305	7			Trapping Rain Water (Prefix Max / Min)	Link	★★★★★				
306	8			Range Sum Query 2D - Immutable	Link	★★★★★				
307	9			Matrix Block Sum	Link	★★★★★				
308	10			Maximum Sum of a 2D Subgrid of size K x K	Link	★★★★★				
309	1			Move Zeroes (Two Pointer Intuition)	Link	☆☆☆☆				

[illegible]

[illegible]

[illegible]